

Chapter 4

4.1

(a) The estimated logistic regression model is: $\text{logit}[\hat{P}(Y = 1)] = -3.777 + 0.1449 \times LI$. The median

effective level, at which $\hat{P}(Y = 1) = 0.50$ is at $-\frac{\hat{\alpha}}{\hat{\beta}} = \frac{3.771}{0.1449} \approx 26$.

(b) Since $\hat{\beta} = 0.1449$, $e^{\hat{\beta}} = e^{0.1449} \approx 1.16$. Thus, when LI increases by 1, the estimated odds of remission multiply by 1.16.

(c) Interquartile range: the 25th quartile and 75th quartile for LI are 12 and 26. $\hat{P}(Y = 1)$ increases from 0.115 to 0.497, between those values.

Range: the minimum and maximum for LI are 8 and 38. $\hat{P}(Y = 1)$ increases from 0.068 to 0.849, between those values.

(d) When $LI = 8$, $\hat{P}(Y = 1) = 0.068$. Thus, the rate of change in $\hat{P}(Y = 1)$ when $LI = 8$ is

$$\hat{\beta}\hat{P}(Y = 1)[1 - \hat{P}(Y = 1)] = 0.14486(0.068)(0.932) = 0.009177 \approx 0.009.$$

(e) Average marginal effect = 0.0226, which means that the average rate of change is 0.0226 in the estimated probability of a cancer patient achieving remission per 1-cm increase in labeling index.

4.3

ML estimates are the same, but deviance is not, because the number of parameters in the saturated model is 27 for ungrouped data but 14 for grouped data.

4.5

(a) The estimated logistic regression model is: $\text{logit}[\hat{P}(Y = 1)] = 15.043 - 0.232x$. When x increases by 1 unit, the estimated odds multiply by $\exp(-0.232) = 0.793$.

(b) When temperature is at 31, the estimated probability is equal to 0.9996.

(c) $\hat{P}(Y = 1) = 0.50$ occurs at temperature = 64.8. The rate of change in $\hat{P}(Y = 1)$ per degree increase in temperature when $x = 64.8$ is $-0.232(0.50)(0.50) = -0.058$.

(d) Estimated odds of thermal distress multiply by $\exp(-0.232) = 0.79$ for each 1-degree increase in temperature.

(e) Wald statistic is $z = -\frac{0.2322}{0.1082} = -2.145$ with p -value = 0.032. Likelihood-ratio statistic is 7.952 with

$df = 1$, the p -value is 0.005. Since the Wald test and likelihood-ratio test are both significant, it implies the

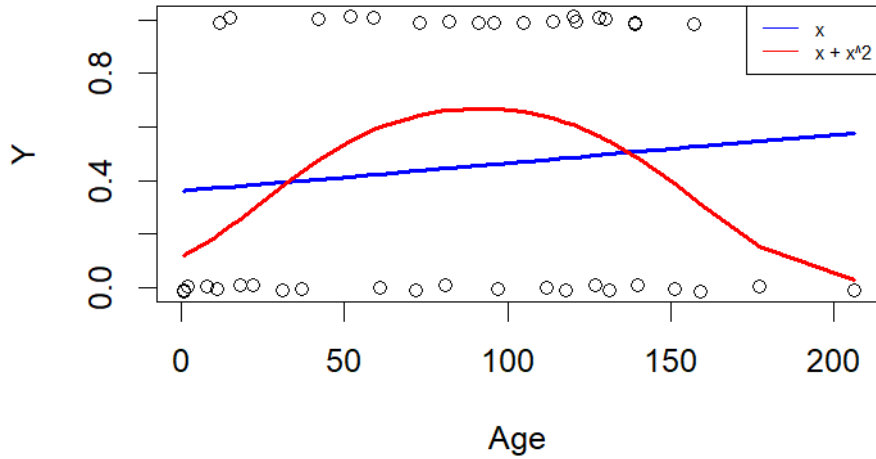
effect of temperature is significant.

4.7

(a) The estimated logistic regression model is: $\text{logit}[\hat{P}(Y = 1)] = -0.573 + 0.0043x$. $z = \frac{\hat{\beta}}{SE} = \frac{0.004296}{0.005849} = 0.734$, the p-value is equal to 0.463, which implies that the age effect is not significant.

(b) Age values are more disperse when kyphosis absent.

(c) The estimated logistic regression model is: $\text{logit}[\hat{P}(Y = 1)] = -2.046 + 0.06x - 0.0003x^2$. The Wald statistic is $z = -\frac{0.0003279}{0.0001564} = -2.097 < z_{0.025} = -1.96$, the LR statistic is $54.504 - 48.228 = 6.276 > \chi^2_{0.05,1} = 3.84$. Since the Wald test and likelihood-ratio test are both significant, it implies the effect of squared age term is significant.



4.9

(a)

$\text{logit}[\hat{P}(Y = 1)] = 1.0986 - 0.1226c_2 - 0.7309c_3 - 1.8608c_4$ (reference = 1), or

$\text{logit}[\hat{P}(Y = 1)] = -0.7621 + 1.8608c_1 + 1.7382c_2 + 1.1299c_3$ (reference = 4).

The estimated odds a medium light crab has a satellite are $e^{1.86} = 6.4$ times estimated odds a dark crab has a satellite.

(b) The likelihood-ratio test statistic is 13.698, which follows chi-square distribution with $df = 3$. The p-value = 0.003, which implies the effect of color is significant.

(c) $\text{logit}[\hat{P}(Y = 1)] = 2.3635 - 0.7147c$. Estimated odds of having a satellite multiply by $e^{-0.7147} = 0.49$

for each 1 unit increase in color. $z = \frac{\hat{\beta}}{SE} = -\frac{0.7147}{0.2095} = -3.412$, the p-value is equal to $0.000645 < 0.05$.

(d) Advantage of treating color as quantitative is that the test of the effect is more powerful, but the disadvantage is that there may not be linear trend for color effect.

(e) The estimated logistic regression model is: $\text{logit}[\hat{P}(Y = 1)] = -2.0316 + 1.6531w - 0.5142c$.

$s_w = 0.577$ and $s_c = 0.802$.

The standardized coefficient of weight is: $1.6531 \times 0.577 = 0.954$.

The standardized coefficient of quantitative color is: $-0.5142 \times 0.802 = -0.412$.

A standard deviation increase in weight has more than double the effect of a standard deviation increase in color, adjusting for the other variable.

4.17

(a)

$r = 1$: $\text{logit}[\hat{P}(Y = 1)] = -6.7 + 0.1a + 1.4s$. $r = 0$: $\text{logit}[\hat{P}(Y = 1)] = -7.0 + 0.1a + 1.2s$.

The conditional odds ratio for the smoking effect is $\exp(1.4) = 4.1$ for blacks ($r = 1$) and $\exp(1.2) = 3.3$ for whites ($r = 0$).

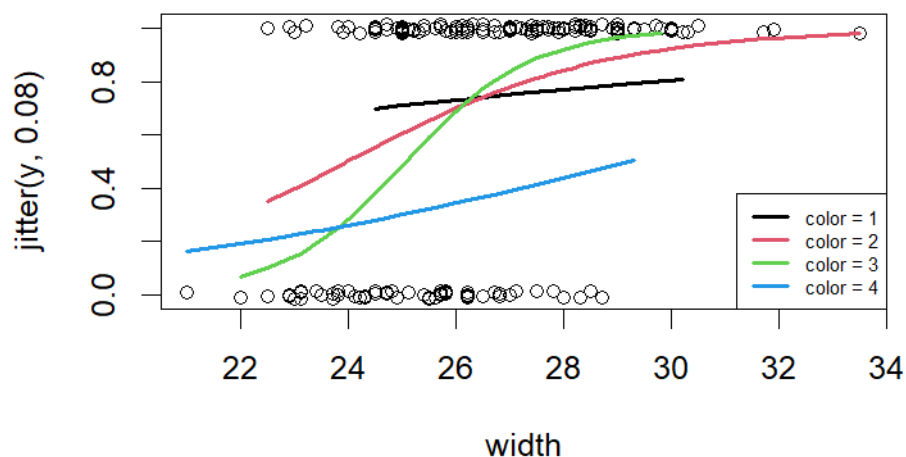
$s = 1$: $\text{logit}[\hat{P}(Y = 1)] = -5.8 + 0.1a + 0.5r$. $s = 0$: $\text{logit}[\hat{P}(Y = 1)] = -7.0 + 0.1a + 0.3r$.

The conditional odds ratio for the race effect is $\exp(0.5) = 1.649$ for smokers ($r = 1$) and $\exp(0.3) = 1.350$ for nonsmokers ($r = 0$).

(b) The coefficient of s of 1.2 is log odds ratio between y and s when $r = 0$ (whites), in which case interaction does not enter the equation. The coefficient of r of 0.3 is log odds ratio between y and r when $s = 0$ (whites), in which case interaction does not enter the equation. The P -value represents result of test that log odds ratio between y and s for $r = 0$ and between y and r for $s = 0$.

4.19

(a)



$\text{logit}[\hat{P}(Y = 1)] = -1.753 - 8.287 c_2 - 19.765 c_3 - 4.101 c_4 + 0.106x + 0.313(c_2x) + 0.752(c_3x) + 0.094(c_4x)$ (reference = 1), or

$\text{logit}[\hat{P}(Y = 1)] = -5.854 + 4.101 c_1 - 4.186 c_2 - 15.664 c_3 + 0.200x - 0.094(c_1x) + 0.218(c_2x) + 0.658(c_3x)$ (reference = 4).

(b) The likelihood-ratio statistic is 4.38 with $df = 3$, the p - value = 0.2236. There is no sufficient evidence to claim that the interaction model fits better than the simpler model without interaction terms. The correlation between the observed and fitted values for each model is $R = 0.452$ for simpler model and $R = 0.472$ for interaction model, the simpler model fits nearly as well and is adequate according to the test.

4.21

We predict $y = 1$ whenever $\hat{P}(Y = 1) > 0.50$. Of the cases with $y = 1$, they are classified as correct or incorrect according to whether $\hat{P}(Y = 1) > 0.50$ or not. Of the cases with $y = 0$, they are classified as correct or incorrect according to whether $\hat{P}(Y = 1) < 0.50$ or not. Sensitivity = $96/111 = 0.865$, specificity = $31/62 = 0.50$.